

CommuteIQ AI Predictive Commute Intelligence Platform

Reimagining Urban Mobility — Daily Commute in India 2026

Predictive Multi-Modal Journey Planner with Smart Last-Mile Carpooling
Track: Smart Multi-Modal Journey Planner

Team Name: The Innovengers
Members:

1. Vatsal Goyal
2. Rahul Sharma
3. Ishaan Verma
4. Arman Singh

One Destination. Four Smart Strategies.

Introduction

Urban Mobility Challenges:



Traffic Congestion

Gridlock wastes millions of commuter hours every day across Indian cities.



Crowded Transport

Overcrowded buses and metros with no visibility into crowd levels before boarding.



Last-Mile Gap

The final 1–3 km remains the most inefficient, expensive and unplanned stretch.



Fragmented Apps

Commuters juggle multiple disconnected apps with no unified intelligence layer.



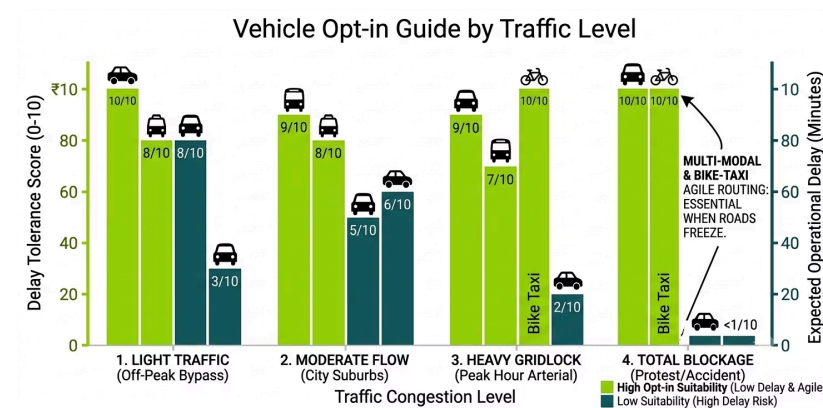
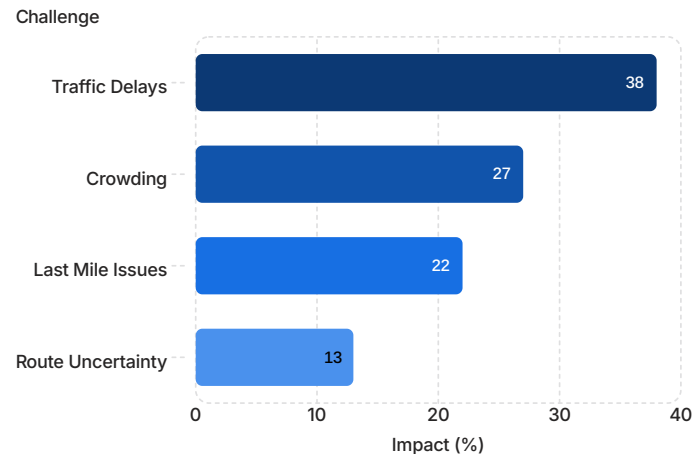
Unpredictable ETAs

No reliable way to know if a route will be delayed before it happens.



1.5–2 Hours in Transit

Many commuters spend 1.5–2 hours daily in transit, leaving less time for work, family, and rest.



Problem Statement

Why Current Journey Planners Are Not Enough?

Today's platforms provide:

-  Routes
-  ETA
-  Navigation

But fail to answer:

 Which route will be **least crowded?**

 Which route is **most reliable?**

 Should I **leave now or later?**

 Which route balances **cost, comfort and time?**

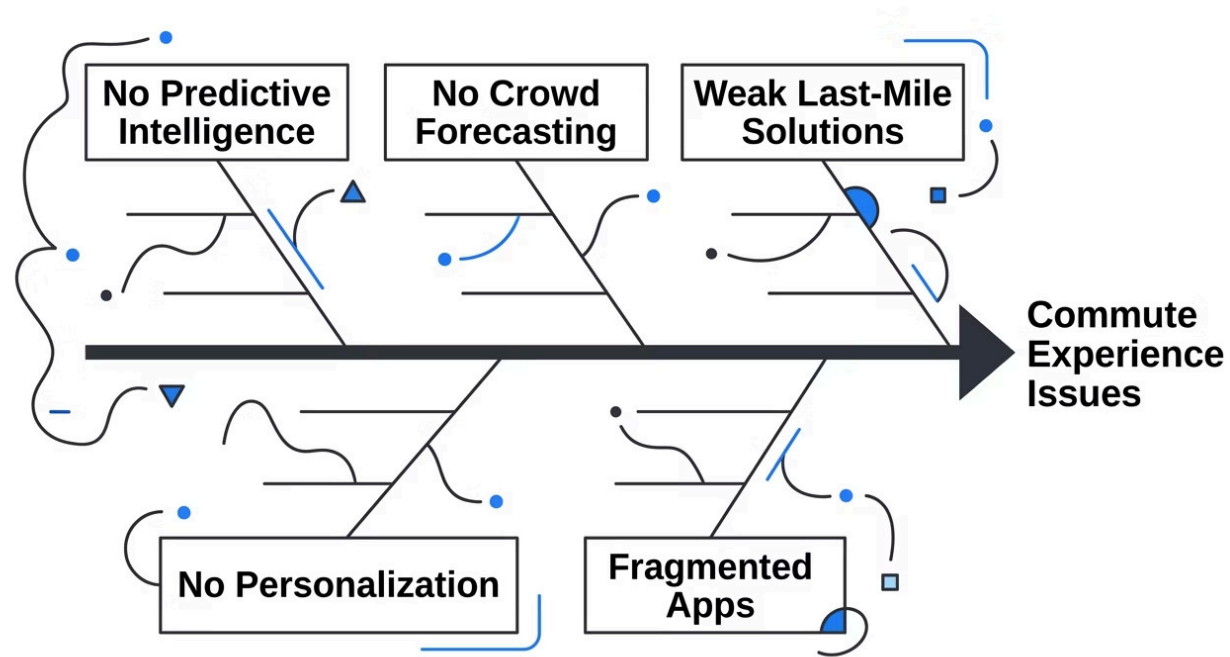
 What is the **best last-mile option?**

 Can I save time by leaving 20 minutes later?

Existing systems optimize *routes*. We optimize the **commuting experience**.

Problem Analysis

Root Causes:



Gap 1 — No Predictive Intelligence

Apps react to traffic. They don't predict future conditions.

Gap 2 — No Crowd Forecasting

Users enter overcrowded buses and metros blindly.

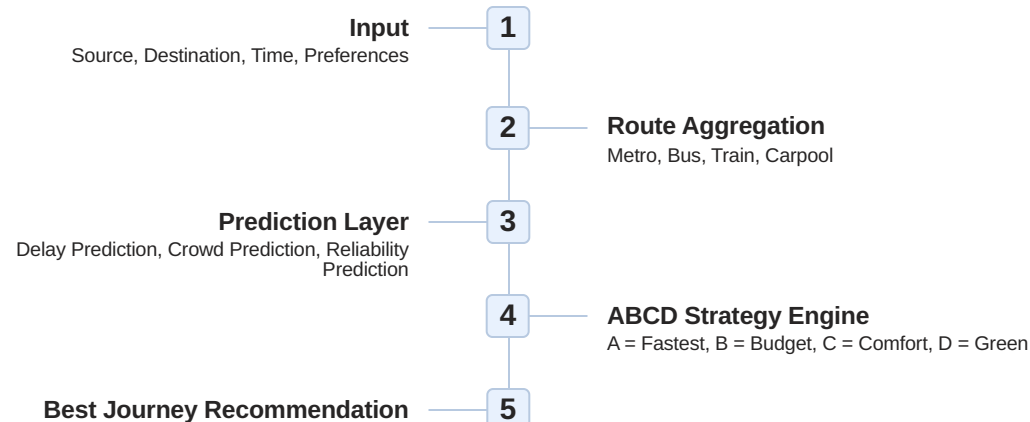
Gap 3 — Weak Last-Mile

Final 1–3 km remains inefficient and unplanned.

Gap 4 — No Personalization

Everyone receives nearly identical recommendations.

Smart Multi-Modal Journey Planning Engine



ABCD Strategy Engine

A = Fastest | B = Budget | C = Comfort | D = Green



Delay Prediction

Predict future delays before they happen using ML models.



Crowd Prediction

Forecast crowd density across metro, bus and train in real time.



Smart Last-Mile Carpool

Intelligent shared ride matching for the final kilometre gap.



Personalized Learning

Continuously learns commuter preferences to sharpen recommendations.



Commute Time Machine

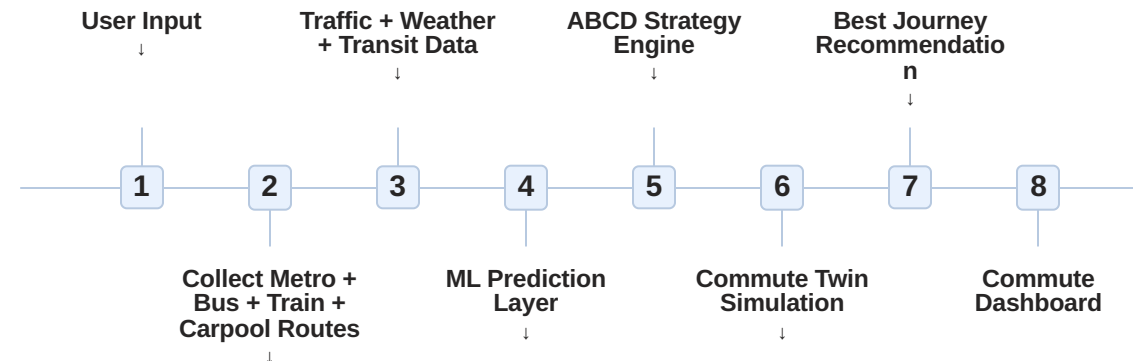
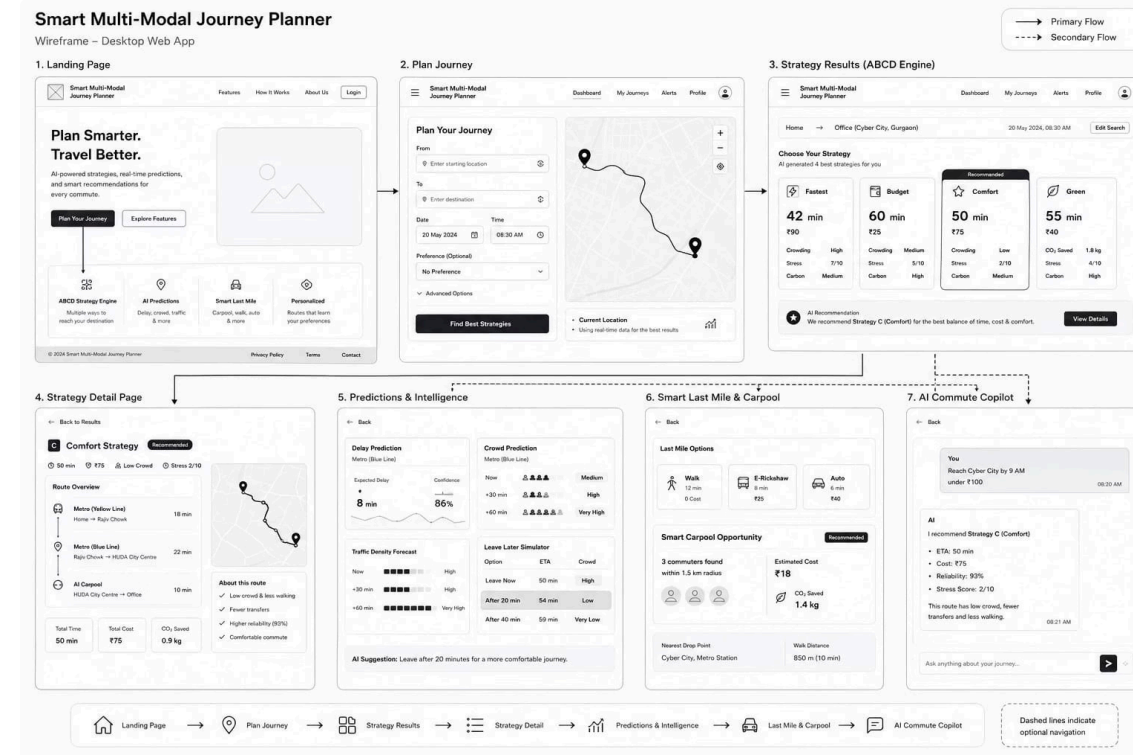
Predicts the best departure time before you travel to minimize delays and optimize your journey.



Core Insight: Not another route planner — a **Predictive Commute Intelligence Platform** that optimizes the full experience.

Workflow

How CommuteIQ AI Works:



Tech Stack

Built on a Modern, Scalable Stack:

Frontend

React.js · Tailwind CSS

Backend

Node.js · Express.js

Database

MongoDB Atlas

Machine Learning

Python · Scikit-Learn · Pandas · NumPy

APIs

Google Maps · OpenWeather · Delhi Open Transit · GTFS Feeds

Architecture:

React Frontend

Interactive UI built with React and Tailwind.

MongoDB Database

Atlas-hosted document store for app data.

Prediction & Strategy

Prediction engine feeding ABCD strategy.

Delay Prediction

Estimates potential disruptions across routes.

Node.js Backend

API layer using Express and scalable services.

ML Service (Python)

Data processing with Pandas and Scikit-Learn.

Forecasting

Predictive planning for upcoming commute conditions.

Reliability Scoring

Rates route confidence for better recommendations.

User Impact

Impact Across Every Dimension of the Commute:



Efficiency

Reduced travel time through predictive routing and real-time delay avoidance.



Cost Savings

Optimized route selection across budget-friendly multi-modal options.



Sustainability

Reduced emissions by promoting metro, public transport and carpooling.



Convenience

One platform. One recommendation engine. No more app-juggling.



Safety

Lower crowd exposure and more reliable route planning for every commuter.

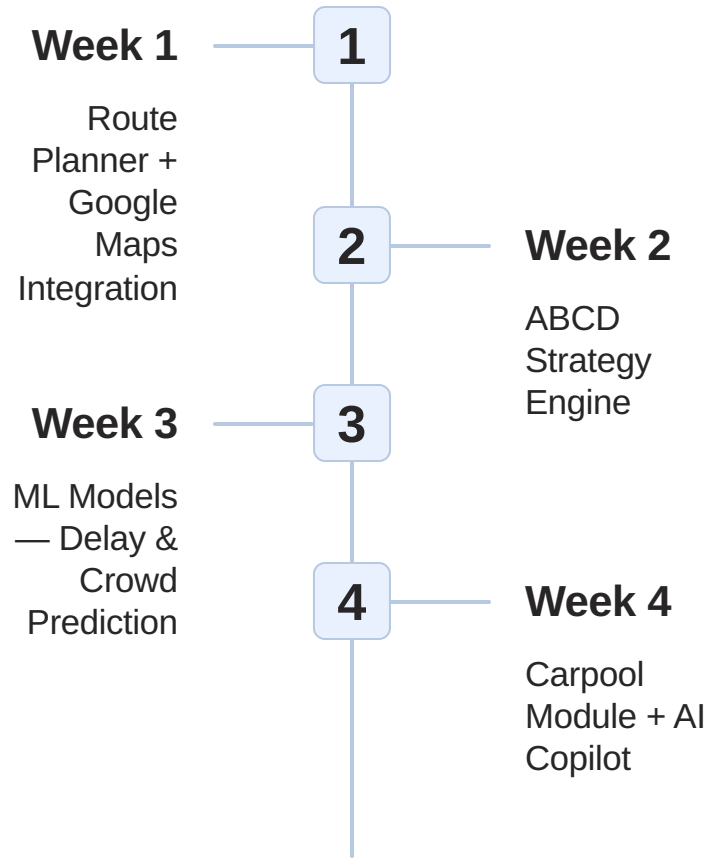
Expected Impact Metrics:

Travel Time	-20%
Commute Cost	-15%
Carbon Emissions	-25%
Route Reliability	+30%

Feasibility

Development Roadmap, Scalability & Risks:

4-Week Roadmap



Scalability Path



Risk & Mitigation

Limited Transit Data

Use public datasets & open APIs

API Costs

Student credits & free tiers

Prediction Accuracy

Continuous model training & feedback

Meet the Team



Vatsal Goyal (Team Lead)

· Backend & AI Systems



Rahul Sharma

· Frontend Development



Ishaan Verma

· Machine Learning & Data



Arman Singh

· UI/UX Design · Documentation

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